MEO DEV â¢ BI KIP GIT VAO BO CHO SINH VIEN IT (TRANG 1/6)

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TAI LIEU DOC QUYEN DANG KY BAN QUYEN BOI MEO DEV AI â¢ GIA TRI: 25.000D TIEN DIEN SINH TON

Loi cam on: Cam on ban da tiep mau giup Meo Dev khong bi lenh rm -rf / xoa so khoi server!

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LOI MO DAU TU SENIOR TECH LEAD:

Chao con sen IT! Nguoi ta bao hoc Dai hoc 4 nam co 2 noi so lon nhat: mot la thi lai Triet hoc, hai la conflict Git lam mat code do an tot nghiep ngay dem truoc gio bao ve.

Tai lieu nay khong day ban ly thuyet han lam. Tao viet ra tap tai lieu nay bang mau va nuoc mat cua hang tram sinh vien da tung bi xoa repo, bi mat commit va bi dong doi "phong sat".

MUC LUC TAI LIEU (6 PHAN CHUYEN SAU):

- Phan 1: Bang tra cuu sieu toc 25 lenh Git hang ngay (Trang 2)
- Phan 2: Bi kip xu ly Merge Conflict phut 89 khong mat 1 dong code (Trang 3)
- Phan 3: Chieu thuc Git Stash cat code tam thoi & Git Rebase gop commit (Trang 4)
- Phan 4: Cap cuu khi lo tay commit sai: Git Reset, Revert va Reflog (Trang 5)
- Phan 5: Quy trinh Git Flow chuan do an nhom 10 diem & Top 10 loi ngau ngo nhat (Trang 6)

Hay luu file PDF nay vao may va mo ra moi khi ban gap su co tren Terminal. Chuc ban vao bo!

MEO DEV â¢ BANG TRA CUU 25 LENH GIT THUC CHIEN (TRANG 2/6)

NHOM 1: KHOI TAO & DONG BO LOCAL - REMOTE

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1. git init : Khoi tao repo local moi.
2. git clone <url> : Clone du an ve may.
3. git remote add origin <url> : Lien ket thu muc local voi GitHub repo.
4. git remote -v : Kiem tra danh sach duong dan remote fetch va push.
5. git fetch origin : Lay danh sach thay doi tu remote ve nhung chua merge.
6. git pull --rebase origin main : Keo code moi va dat commit cua minh len tren cung (cuc sach).

NHOM 2: STAGING & LUU TRU SNAPSHOT

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7. git status -s : Xem trang thai ngan gon (M: modified, A: added, ?: untracked).
8. git add -p : Chon tung doan code (hunk) de stage, khong phai add ca file!
9. git diff : Xem su thay doi cua code chua stage.
10. git diff --staged : Xem su thay doi cua nhung file da dua vao Staging Area.
11. git commit -m "[type]: msg" : Commit chuan Conventional (feat:, fix:, chore:, docs:).
12. git commit --amend -m "[new]" : Sua lai message commit gan nhat neu lo tay go sai chinh ta.

NHOM 3: QUAN LY NHANH (BRANCHING)

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13. git branch : Liet ke cac nhanh local.
14. git branch -M main : Doi ten nhanh mac dinh thanh main (thay the cho master cu).
15. git checkout -b feat/login : Tao va chuyen sang nhanh moi chi bang 1 lenh duy nhat.
16. git switch -c feat/login : Cu phap moi tu Git 2.23+ chuyen nhanh truc quan hon.
17. git branch -d <ten_nhanh> : Xoa nhanh an toan sau khi da merge xong.
18. git branch -D <ten_nhanh> : Ep xoa nhanh bat chap chua merge (dung can than!).

MEO DEV â¢ XU LY MERGE CONFLICT PHUT 89 (TRANG 3/6)

NGUYEN NHAN CONFLICT: Hai nguoi cung sua tren cung mot dong hoac cung mot doan code!

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1. NHAN DIEN CONFLICT MARKERS TRONG FILE CODE:

Khi mo VS Code hoac text editor, ban se thay 3 vach ngan do Git tao ra:

```
<<<<<<< HEAD (Current Change - Code hien tai tren may cua ban)
const API_URL = "https://api.staging.meodev.com";
=======
const API_URL = "https://api.production.meodev.com";
>>>>>>> feat/payment (Incoming Change - Code tu nhanh merge vao)
```

2. QUY TRINH 4 BUOC GIAI QUYET CONFLICT AN TOAN 100%:

- Buoc 1: Binh tinh, goi dien hoac chat ngay voi dong doi nguoi vua commit doan code do.
- Buoc 2: Chon phuong an dung: Giu code cua minh, giu code cua ban, hay ket hop ca hai.
- Buoc 3: Xoa toan bo cac ky tu <<<<<<<, =======, >>>>>>> va luu file lai.
- Buoc 4: Chay chuoi lenh hoan tat merge:

```
git add .
git commit -m "fix(merge): resolve conflicts between main and feat/payment"
git push origin <branch_cua_ban>
```

[WARN] TRUONG HOP XAU NHAT: Khong biet xu ly the nao, muon quay ve truoc khi merge:

Chay ngay lenh: ‘git merge --abort‘ hoac ‘git rebase --abort‘.

Toan bo trang thai repository se quay tro lai nguyen ven nhu chua he co gi xay ra!

MEO DEV â¢ TUYET KY GIT STASH VA GIT REBASE (TRANG 4/6)

A. GIT STASH: CHIEC TUI BAO BOI CAT CODE KHAN CAP

Kich ban: Ban dang code chuc nang A dang do (code con bug, chua the commit). Sep bao: "Pull gap code moi tu main ve de fix bug san xuat". Neu pull se bi conflict hoac Git chan!

- Lenh 1: ‘git stash save "code dang lam do chua test"‘
  -> Toan bo code sua doi bien mat khoi working tree, tra repo ve trang thai sach se.
- Lenh 2: ‘git pull origin main‘ -> Keo code moi ve may muot ma khong loi.
- Lenh 3: ‘git stash pop‘ -> Loi lai code dang lam do ra va tiep tuc chien dau!
- Lenh 4: ‘git stash list‘ -> Xem danh sach toan bo cac goi stash da tung cat.
- Lenh 5: ‘git stash drop stash@{0}‘ -> Xoa bo mot goi stash cu khong dung den nua.

B. GIT REBASE: LAM SACH LICH SU COMMIT TRUOC KHI NOP DO AN

Kich ban: Nhanh cua ban co 10 commit rac: "fix", "fix tiep", "fix bug troi oi", "chay duoc roi".
Giao vien huong dan nhin vao se danh gia kem chuyen nghiep!

- Lenh: ‘git rebase -i HEAD~4‘ (Gop 4 commit gan nhat)
- Git se mo trinh soan thao text. Dong dau tien giu chu ‘pick‘, cac dong con lai sua thanh ‘squash‘ (hoac ‘s‘).
- Luu lai -> Git se gop toan bo 4 commit thanh 1 commit duy nhat voi noi dung chi tiet, sach dep 100%!

MEO DEV â¢ CAP CUU KHI LO TAY COMMIT SAI (TRANG 5/6)

A. PHAN BIET GIT RESET VA GIT REVERT (KHONG DUOC DUNG NHAM!)

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1. ‘git reset --soft HEAD~1‘ (AN TOAN NHAT):
   - Xoa commit gan nhat, nhung GIU NGUYEN toan bo code cua ban o Staging Area.
   - Dung khi: Ban lo tay commit ma quen chua add 1 file quan trong hoac viet sai commit message.

2. ‘git reset --hard HEAD~1‘ (NGUY HIEM CAP DO 1):
   - Xoa commit dong thoi XOA SACH toan bo code da viet o commit do.
   - Chi dung khi: Ban thuc su muon vut bo toan bo nhung gi vua lam.

3. ‘git revert <commit_id>‘ (CHUAN CHINH KHI DA PUSH LEN REMOTE):
   - Tao mot commit moi co noi dung nguoc lai de triet tieu commit sai ma KHONG XOA LICH SU.
   - Khuyen dung tuyet doi khi lam viec nhom, tranh lam vo lich su commit cua thanh vien khac.

B. GIT REFLOG: CO MAY THOI GIAN CUU CODE DA MAT

Neu ban lo tay chay ‘git reset --hard‘ va nghi rang code minh viet 3 ngay qua da bi xoa vinh vien:
- Buoc 1: Chay lenh ‘git reflog‘ -> Git luu giu moi nhat cu nhat dong di chuyen cua HEAD!
- Buoc 2: Tim dong commit truoc khi bi reset (VD: ‘HEAD@{2}‘).
- Buoc 3: Chay ‘git reset --hard HEAD@{2}‘ -> Toan bo code song lai nhu mot phep mau!

MEO DEV â¢ GIT FLOW DO AN NHOM VA TOP LÆ» I (TRANG 6/6)

A. MO HINH GIT FLOW CHUAN CHO NHOM SINH VIEN 3 - 5 NGUOI

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1. Nhanh ‘main‘ (hoac ‘master‘): Chi chua code da test, demo ngon lanh. BAT KHOA BRANCH PROTECTION.
2. Nhanh ‘develop‘: Nhanh trung gian tich hop code cua tat ca thanh vien.
3. Nhanh ca nhan: Dat ten theo quy uoc: ‘feat/ten-chuc-nang‘ hoac ‘fix/ten-bug‘.
   Vi du: ‘feat/shopping-cart‘, ‘feat/auth-jwt‘, ‘fix/login-crash‘.
4. Quy trinh: Code xong tren local -> Push len GitHub -> Tao Pull Request vao ‘develop‘ ->
   Nhom truong review xong moi duoc phep bam Merge.

B. TOP 5 LOI NGAO NGO SINH VIEN IT HAY BI DANH TRUOT:

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1. Commit ca thu muc ‘node_modules/‘: Lam repo nang 500MB -> Khac phuc: Luon co file ‘.gitignore‘!
2. Commit file ‘.env‘ chua API Key / DB Password: Bi hacker quet bot rut sach tien Cloud ->
   Khac phuc: Them ngay ‘.env*‘ vao ‘.gitignore‘ va dung ‘.env.example‘.
3. Dung lenh ‘git push --force‘ len main lam de commit cua dong doi -> CUC KY TOI KY!
4. Dinh trang thai "Detached HEAD" do checkout commit cu -> Khac phuc: ‘git switch -‘ de ve nhanh cu.
5. Dat commit message vo nghia: "a", "asdfgh", "chay di ma" -> Thay bang Conventional Commit chuan.

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CHUC BAN HOAN THANH XUAT SAC DO AN! MEO DEV CHUC BAN RA TRUONG LUONG 30 CU! â¢ MEOW! ð ¾

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